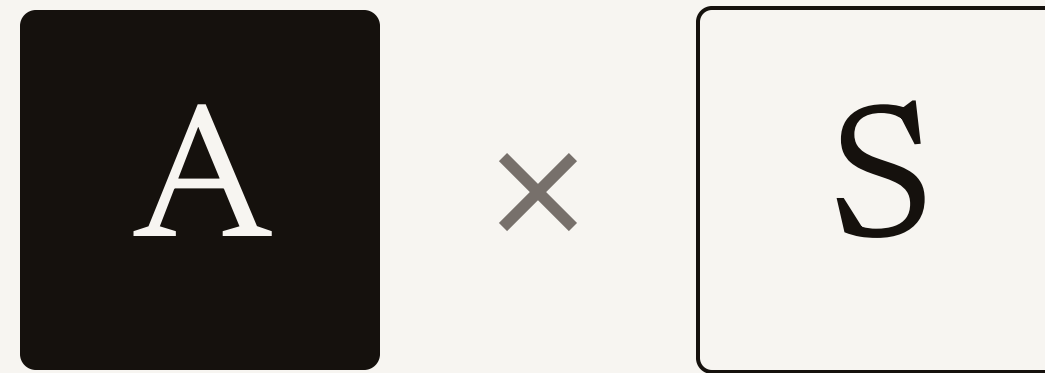




Apple vs. Samsung

A Strategic Economic Analysis



Apple won in court. *Samsung won the market.*

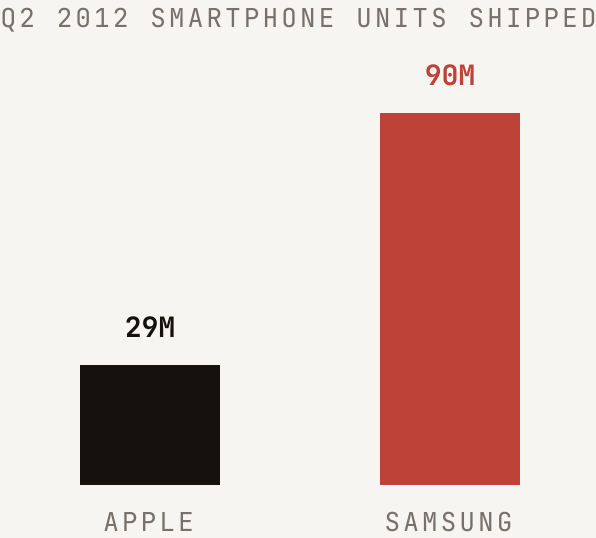
How can both be true?



AUGUST 2012 – VERDICT

\$1.05B

LARGEST PATENT VERDICT IN U.S. HISTORY



FROM QUESTION TO VERDICT

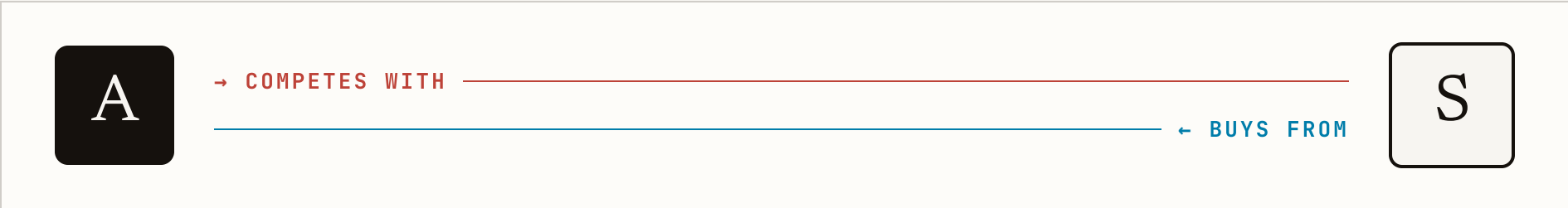
Our argument in *six beats*.

I	The Setup Why they couldn't just walk away.	SLIDE 5
II	Why Apple Sued Bargaining failure, three pressures, expert critique.	SLIDES 7-12
III	The Pivot Why the threat had a ceiling.	SLIDE 13
IV	The Logic Two parallel games at once.	SLIDES 14-15
V	The Evidence What investors actually saw.	SLIDES 16-17
VI	The Verdict Who won — and what it means.	SLIDE 18

I · ONE STRANGE RELATIONSHIP

Rivals *and* supplier–customer at the same time.

This duality — **coopetition** — constrains every move that follows in the dispute.



FACT	NUMBER
Samsung's share of iPhone manufacturing cost	26%
Apple's component purchases from Samsung (2010)	\$6B
Apple's share of Samsung's total revenue	4%

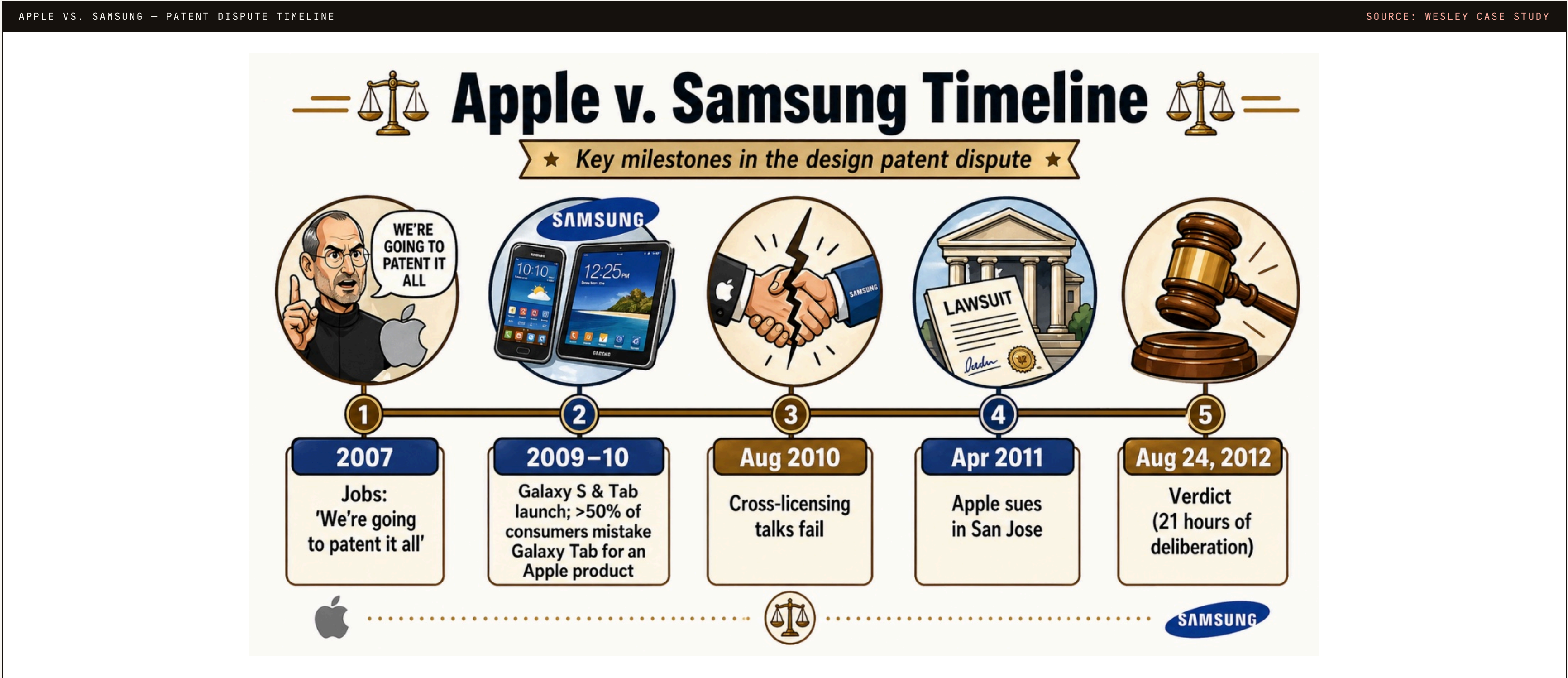
"Apple was reluctant to take legal action against Samsung, given its key role in supplying chips and components."

— WESLEY CASE STUDY

SEVEN YEARS IN FIVE BEATS

From Jobs's promise to the largest patent verdict in U.S. history.

2007 → 2012
5 BEATS • 1 VERDICT



BEFORE THE THREE PRESSURES • THE AUG 2010 COLLAPSE

Five reasons the deal couldn't close.

01

A ONE-WAY TAX, NOT A TRADE

Jobs offered Samsung an Apple license at **~\$30/phone + ~\$40/tablet** ($\approx$ \$250M/year per the trial record). It was a license fee — Apple did not propose to pay for Samsung's IP in return. *A one-way license is not a cross-license.*

02

ASYMMETRIC PORTFOLIOS

Apple's strong patents were **UI & design** — the prestige stack, premium-priced. Samsung's were **3G wireless SEPs**, legally capped by FRAND duty. The two stacks couldn't trade at par.

03

DESIGN PATENTS DON'T CROSS-LICENSE

Utility patents license a *function* — both sides can still look different. Design patents license *identity*. Trading the iPhone's look would **erase the very premium Apple was suing to protect.** (Hotelling – slide 10)

04

CREDIBILITY AGAINST OTHER DEFENDANTS

Apple was simultaneously suing **HTC, Motorola, Nokia, Amazon, Google**. Settling cheaply with Samsung would have collapsed every other threat. Refusing to license was Apple's only credible signal to all of them. (see slide 13)

05

NO AGREED VALUATION

Each side believed its own portfolio was the stronger. With no neutral way to price the patents against each other, no deal converges — the *classic cross-licensing failure mode*.

The standard fix doesn't work when the patents protect identity, not function.

– WHY SUING WAS APPLE'S ONLY LEVER LEFT

II • APPLE DID NOT SUE OUT OF ANGER

Three economic forces pushed Apple *into court*.

Each one is the subject of one of the next three slides.

1 PRESSURE #1

Patent Hold-Up

Too many patents, no clean way out — cross-licensing had already failed.

2 PRESSURE #2

Lost Differentiation

Samsung looked too much like Apple — and Apple's premium was eroding fast.

3 PRESSURE #3

Platform Tipping

Android was tipping the OS market — and second place might not survive.

IF YOU REMEMBER NOTHING ELSE FROM THIS DECK — REMEMBER THESE THREE.

WHY APPLE SUED – I

When no one owns the road, everyone pays a toll.

~10K patents needed
TO SHIP ONE SMARTPHONE

\$10B / year, industry-wide
ON PATENT LITIGATION BY 2011

Apple & Google each spent *more on lawsuits than R&D*.
– ECEB SECTION 3.1

Cross-licensing is the textbook fix. It failed in Aug 2010.

PATENT STRATEGY – COMPARED

	APPLE	SAMSUNG
Offensive	Sue Samsung for copying iPhone	Counter-sue over wireless patents
Defensive	"Patent everything" — wall of patents	Build own patent stockpile
Bargaining chip	Win cases → better licensing terms	Wireless patents as leverage

WHY APPLE SUED - II

Bertrand Competition & Hotelling, in plain sight.

BERTRAND COMPETITION

Identical products → price war → zero profit.

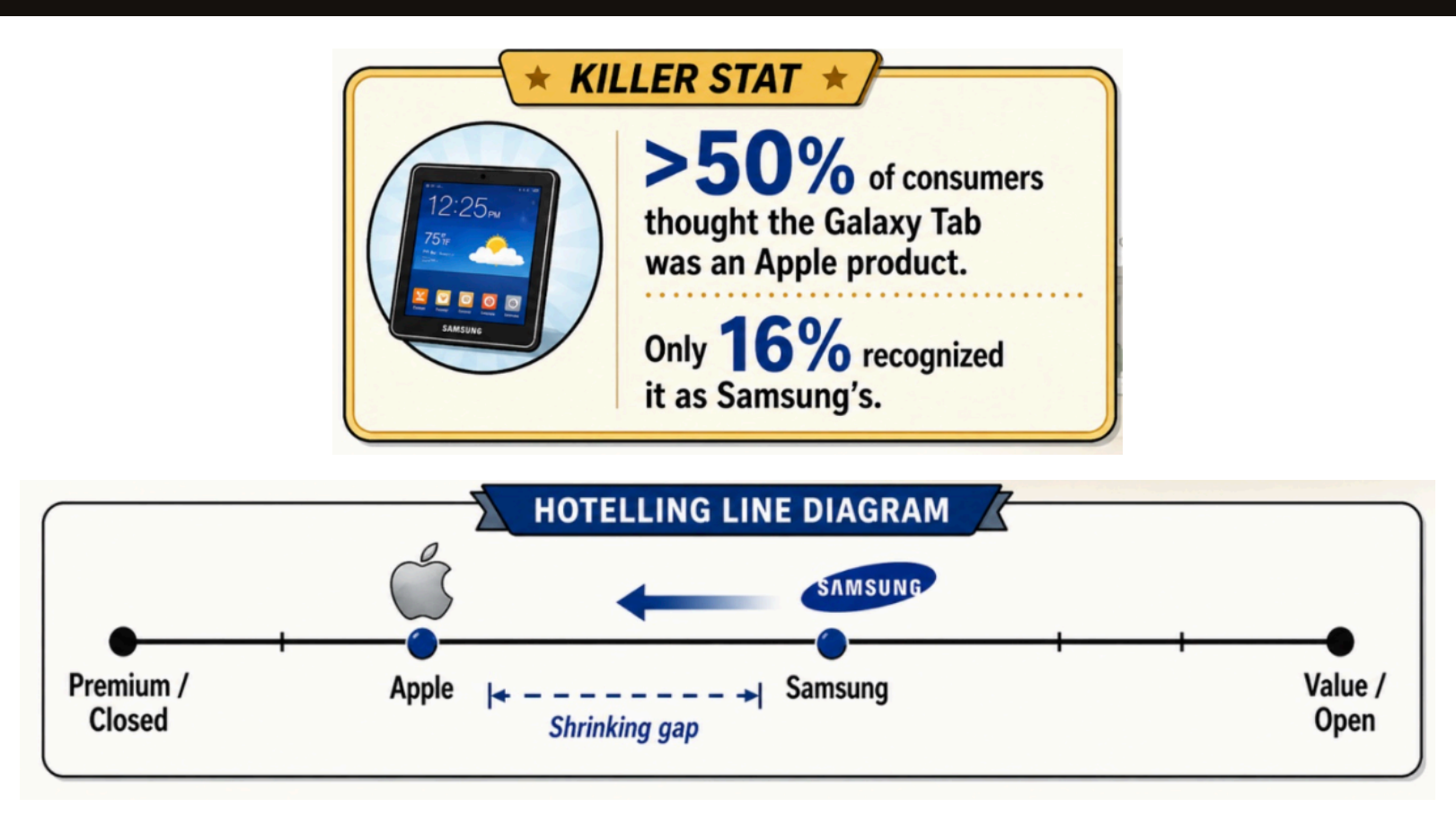
HOTELLING

The closer rivals stand on the "taste line," the harder they compete.

IMPLICATION

Samsung's copying moved it toward Apple → eroded the premium.

FIG. 2 • KILLER STAT + HOTELLING LINE



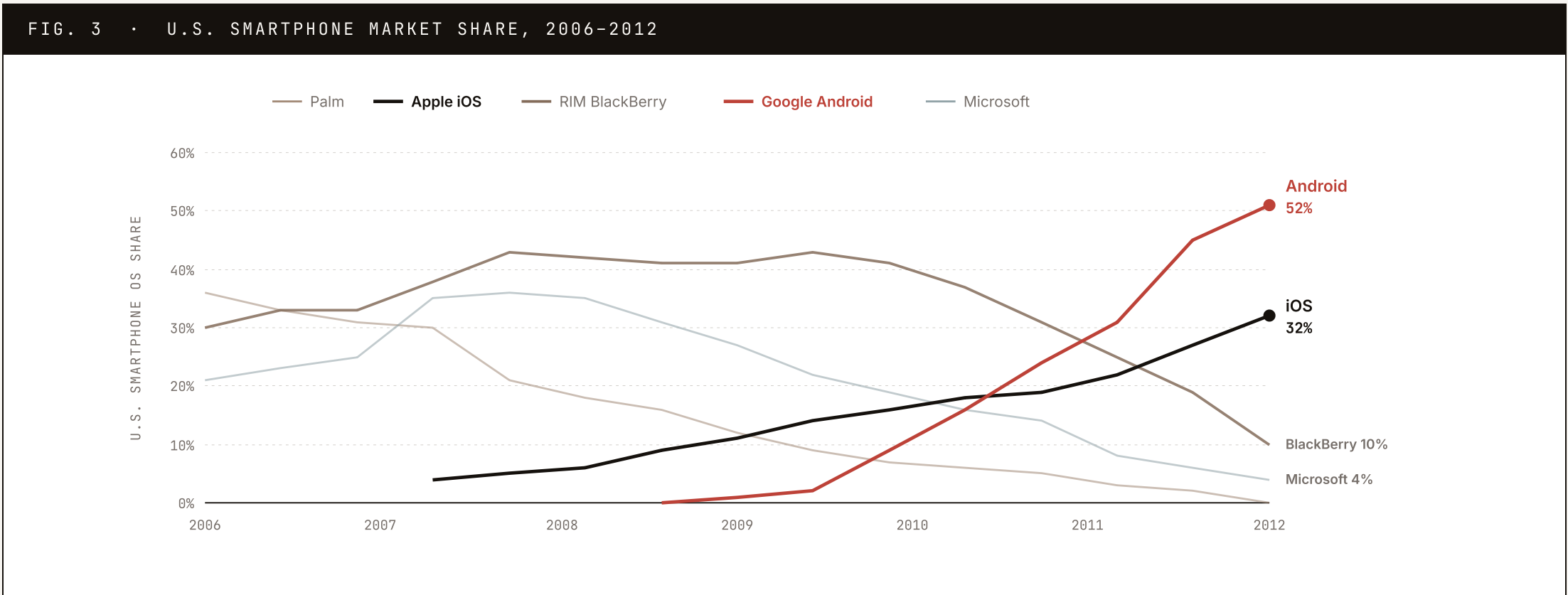
CAPTION: AS SAMSUNG'S POSITIONING APPROACHED APPLE'S, THE DIFFERENTIATION PREMIUM ERODED.

WHY APPLE SUED – III

Network effects don't reward second place.

TWO PLATFORMS, TWO GAMES

	IOS (APPLE)	ANDROID (SAMSUNG+)
Standard	Closed	Open
Network effect	Apps → users	Same — across many OEMs
Switching costs	High (apps, iCloud)	Low between brands
Game	Competes FOR the market	Competes IN the market



SOURCE: CASE EXHIBIT 10 – CANALYS, SILICON VALLEY INSIDER, COMSCORE, GARTNER.

THE VIEW FROM THE BENCH

"Patents are part of the problem."

*"They are not an ideal solution to the problem; instead they are **part of the problem**. Although they are supposed to be limited to inventions that are novel, useful, and non-obvious, they are granted rather **promiscuously** by the Patent and Trademark Office. And while they can be challenged in court, **jurors tend to be biased in favor of patent holders**."*

RICHARD A. POSNER · SLATE, OCT 15, 2012 · CASE P. 2

POSNER'S CLAIM → EVIDENCE FROM THIS CASE

1. PTO grants are promiscuous.
'381 was 62 pages with 20 broad claims; the USPTO later issued a **non-final rejection** of the same patent in 2013.

2. Jurors favor patent holders.
Verdict in **21 hours**; jury upheld every Apple claim despite documented prior art (LG Prada, Knight-Ridder tablet).

3. Patents become weapons, not incentives.
\$10B/yr industry-wide on patent litigation by 2011; Apple and Google each **spent more on lawsuits than R&D** (case p. 2).

4. The system raises the cost of inventing.
Defensive arms race; ~10K patents per smartphone (slide 09) — the system *raises* the cost of inventing in dense fields.

*The aftermath proved Posner right: **AIA inter partes review** (2011) — post-grant patent challenges · **Alice v. CLS Bank** (2014) — software-abstraction limit · **TC Heartland** (2017) — venue reform · **SCOTUS 8–0** in this case (2016) — design-damages narrowed.*

III · THE PIVOT – APPLE'S THREAT HAD A CEILING

"I will spend my last dying breath... every penny of Apple's \$40 billion in the bank... I'm going to destroy Android."

– STEVE JOBS

STRATEGIC COMMITMENT – 3 CRITERIA

Observable?

Public lawsuit + injunctions sought

✓ YES

Understandable?

Copy us → face \$1.05B+ costs

✓ YES

Credible?

Sunk costs, reputation on the line

△ PARTIAL

WHY ONLY PARTIAL?

PUSH TO SUE

Already suing **HTC, Motorola, Nokia, Amazon** and **Google** (Exhibit 2) → softening on Samsung breaks every other threat.

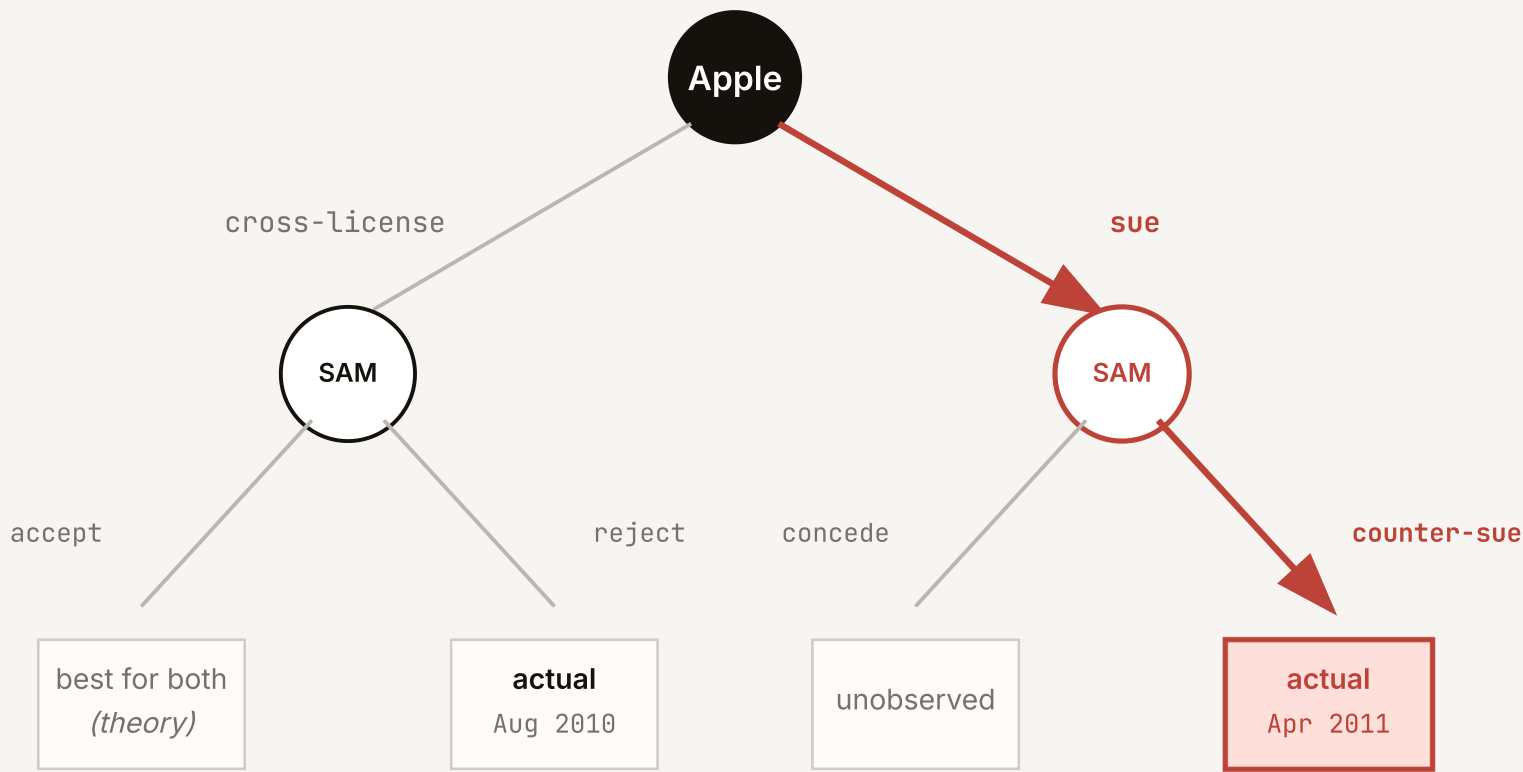
trigger strategy / reputation in a repeated game · suit is the minimum-viable move

PULL TO HOLD BACK

Samsung supplies Apple's chips, screens, memory → can't actually push to ban.

coopetition / hold-up

GAME TREE – HIGHLIGHTED: ACTUAL PATH



CROSS-LICENSE → REJECT → SUE → COUNTER-SUE

APPLE'S THREAT LACKED FULL CREDIBILITY – BOTH SIDES SETTLED IN COURT.

IV · COOPETITION LOGIC

Apple and Samsung play *two games at once*
— and the second one saved the relationship.

GAME 1 Smartphone Competition

PRISONER'S DILEMMA

	SAM: DIFFERENTIATE	SAM: IMITATE
APPLE: TOLERATE	Mod π, Mod π	Low π, High π
APPLE: SUE	–Legal, 0	–Legal, –Legal

GAME 2 Supply Chain Cooperation

REPEATED GAME

	SAM: RELIABLE SUPPLY	SAM: DISRUPT
APPLE: CONTINUE	★ \$6B+, \$6B+ NE	Crippled, Short gain
APPLE: DIVERSIFY	Transition cost, Lose customer	Both lose, Both lose

COOPERATION INEQUALITY

$$r \leq \frac{\pi^C - \pi^{NC}}{\pi^D - \pi^C}$$

r

DISCOUNT RATE

your impatience · ~5–10%

π^C

COOPERATE

\$6B / \$6B · supply & buy continues

π^{NC}

NO COOPERATION

price war wipes profit · **≈ \$0**

π^D

DEFECT ONCE

short-term spike, then relationship collapses

Long-term reward ÷ one-time temptation. With **\$6B/yr** at stake, the inequality holds easily.

WHO INNOVATES, AND WHY

Two effects, two players, two strategies.

REPLACEMENT EFFECT

Apple — the entrant.

Nothing to lose.

Pre-iPhone, Apple had \$0 smartphone revenue.

Big disruptive bet.

The touchscreen iPhone — bet the company on a new category.

EFFICIENCY EFFECT

Samsung — the incumbent.

Now has profits to protect.

Incremental gains: better screens, chips, scale.

R&D climbs incrementally.

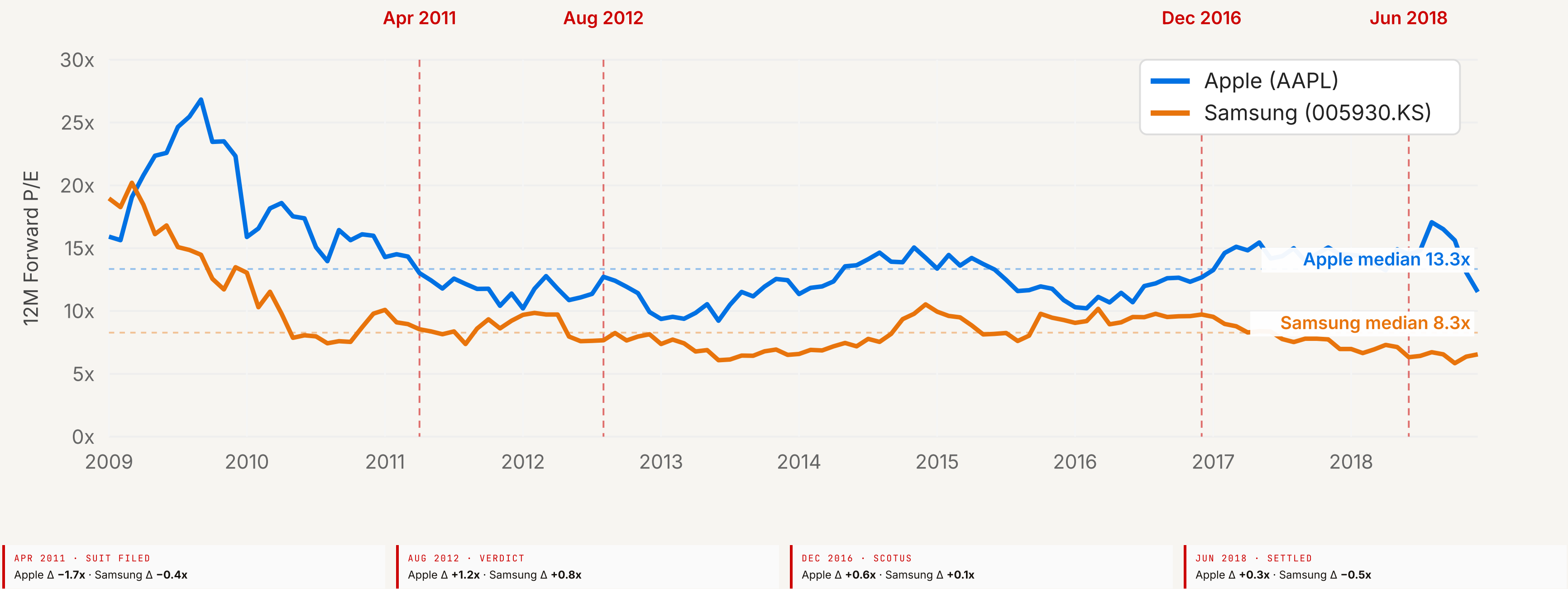
\$8.3B → \$9.1B (2010 → 2011).

THE PRINCIPLE

*"The more you have to lose,
the smaller your bets."*

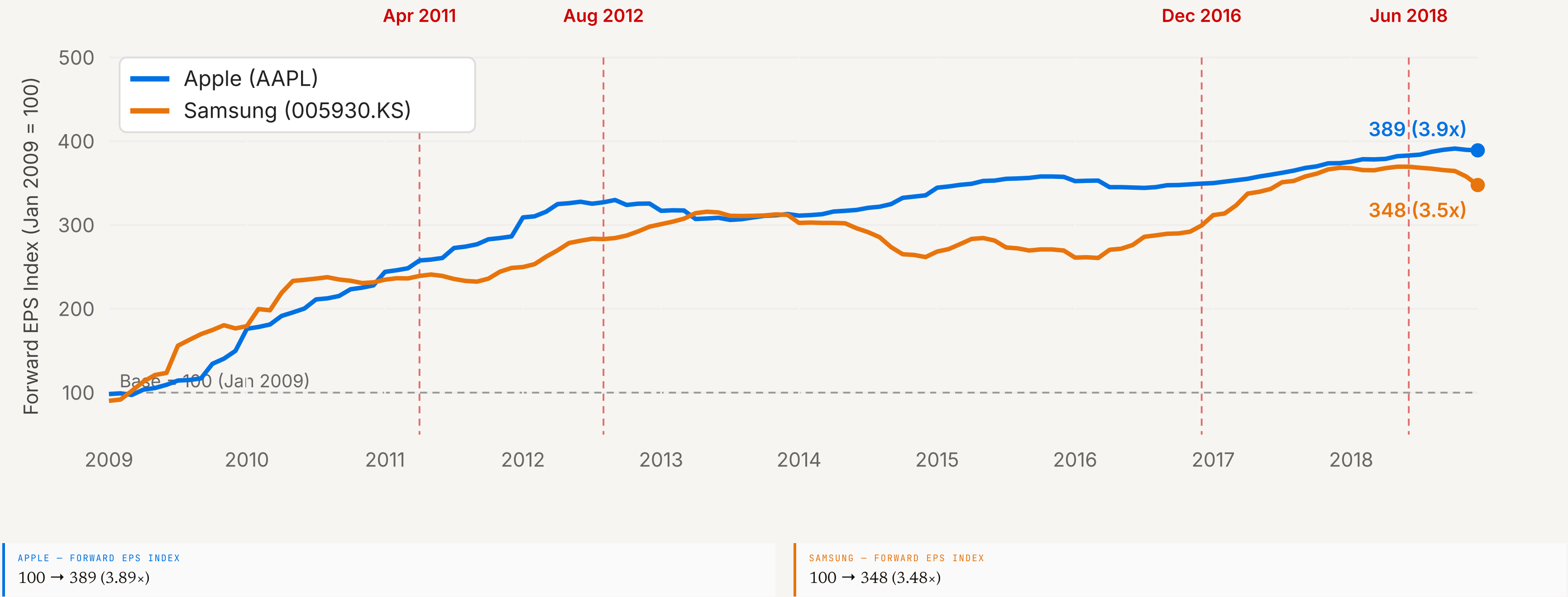
V • WHAT MARKETS SAID

Investors never re-rated either firm on lawsuit news.



WHAT EARNINGS DID

Earnings nearly 4× for both firms — through the entire dispute.



VI • THE VERDICT, FIVE YEARS LATER

Two scorecards. Two stories.

DAMAGES AWARDED

\$539M

8.4% OF SAMSUNG'S 2011 NET INCOME

GOAL OUTCOMES – APPLE VS. SAMSUNG

Apple	
Goal	Outcome
 Damages	\$539M (reduced from \$1.05B) – <i>Partial</i>
 Ban Samsung products	Failed
 Slow Android	Failed (Android hit 52%)

SAMSUNG	
Goal	Outcome
 Become #1 smartphone maker	✓ 90M vs 29M, Q2 2012
 Maintain supply relationship	✓
 Establish Android	✓

CLOSING THOUGHT

“

When companies depend on each other,
patents are bargaining chips, not weapons.

”

Courts move slowly. Markets don't wait.

EPILOGUE I · SAMSUNG'S SCALE

Six years of growth, in USD billions.

METRIC	2006	2007	2008	2009	2010	2011
Sales	89.4	106.0	110.1	106.8	133.8	148.9
R&D	*	*	*	5.8	7.9	9.0
Net Income	8.3	8.0	5.0	7.6	14.0	12.4
Net Margin	9.3%	7.5%	4.6%	7.2%	10.4%	8.3%
Equity	50.2	60.2	57.1	57.2	77.3	91.9

SOURCE: SAMSUNG AUDIT REPORTS. CONVERTED TO USD AT FRB H.10 PERIOD-AVERAGE KRW/USD: 2006 955.5 · 2007 929.3 · 2008 1,102.1 · 2009 1,276.9 · 2010 1,156.1 · 2011 1,108.3. *R&D BUNDLED IN SG&A PRE-2009 (K-GAAP).

EPILOGUE II • FIVE YEARS LATER

Seven years of litigation, quietly settled.

2016

U.S. SUPREME COURT — 8-0

Design-patent damages should be calculated only on the *copied component* (e.g., the front face), not the whole phone — narrowing the \$399M design-patent portion.

2018

RETRIAL & SETTLEMENT

\$539M awarded. A confidential settlement in June 2018 ended seven years of litigation.

& yet

THE QUIET TRUTH

Despite the war, the supply partnership continued.

CASE QUESTION I · MARKET DEVELOPMENT

How this case rewrote the rules of mobile.

01

DUOPOLY CEMENTED

By Q2 2012: **Android 52%** · **iOS 32%**. BlackBerry collapsed 43→4%. Litigation cost-of-entry left no room for a third platform.

02

DESIGN-AROUND CULTURE

Every Android OEM built legal-and-design teams to engineer around Apple's claims — Samsung redesigned the Galaxy Tab 10.1 and pulled its industrial-design language deliberately away from Apple's after 2012. Differentiation became *defensive engineering*, not aesthetic instinct.

03

DAMAGES DOCTRINE NARROWED — SCOTUS 8-0 (2016)

Design-patent damages limited to **the copied component**, not the whole device — a rule change *directly traceable to this case*.

04

FRAND / SEP NORMS TIGHTENED

Samsung's wireless-SEP counter-suit clarified **FRAND** licensing duty — standards-essential patents cannot be priced like premium IP.

Apple won the case. Samsung won the market. The industry rewrote its rulebook.

— Q1 TAKEAWAY

CASE QUESTION II • THE COUNTER AT TRIAL

Strong on paper. Weak in the jury box.

UTILITY PATENT

U.S. #7,469,381 *"bounce-back"*

List scrolling and document translation, scaling, and rotation on a touch-screen display.

SAMSUNG'S GROUNDS (PER JUSTICE KOH)

1 • Anticipation by prior art	primary attack
2 • Obviousness	primary attack

PRIOR ART CITED

DiamondTouch (MERL); Tablecloth (LaunchTile); academic touch-UI demos pre-2007 — all argued to show the elastic-feedback gesture before Apple's filing date.

OUTCOME

Jury upheld validity (2012). **The USPTO issued a non-final rejection of '381 in 2013** on prior-art grounds — Samsung's theory had merit it could not deliver to a jury.

Why Samsung lost despite real evidence — three forces stacked against it:

1. The **clear-and-convincing** burden of proof to invalidate a patent
2. The **ordinary-observer** test rewards consumer confusion
3. Juror bias toward patent holders (see slide 12 – Posner)

DESIGN PATENTS

D'677 • D'087 • D'889

D'677 black face • D'087 rounded rectangle • D'889 tablet form. *Each design patent has a single claim — the drawings themselves.*

SAMSUNG'S GROUNDS

1 • Anticipation by single prior art	primary attack
2 • Obvious to ordinary designer	primary attack

PRIOR ART CITED

LG Prada (2006); Sony "Jony" tablet sketches by Apple-internal designer Nishibori; the 2001 Knight-Ridder tablet concept video for D'889.

OUTCOME

Validity upheld. The *"ordinary observer"* test favored Apple given >50% Galaxy-Tab/iPad consumer-confusion data.

